Can primordial deuterium fix the dark-matter density? DESI DR2 distances read with Ω_c/Ω_b = 3√3 and no CMB, the Hubble constant, $a_0 = cH_0/6$, and a registered test for DESI beyond DR2

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Abstract

Standard big-bang nucleosynthesis, used here as a dictionary of laboratory data, constrains the baryon density ω_b with the deuterium measured in quasar absorbers (and helium, in the primary dictionary). If the dark-matter density is ω_c = 3√3·ω_b, the 13 DESI DR2 BAO distances are read with one cosmological parameter fewer than in flat ΛCDM. No CMB data enter the fits. Results. A premise of the framework on time treats the standard account of the first minutes as a reading in the present. Six translations into measured quantities were registered before computing, and run with PRyMordial (NACRE II rates with LUNA): The predicted lithium stays about 3.5 times the observed value in every reading that holds. Registered test , before the data: the first DESI release beyond DR2, read with the same chain without the CMB. $\Delta\chi^2 \le 4$ «holds», 4-9 «tension», > 9 «the reading 3√3 does not read BAO + BBN». A secondary block compares $a_0 = cH_0/6$ with the first g† measured with a total error $\le 0.06\cdot10^{-10}$. An addendum registered before the data reports the same computation with evolving dark energy next to the main block, with a qualifier fixed in advance. Decision programs with tests are included.

Version 2.58 of the Mathematical Materiality (MM) series, October 2026. It continues MM 2.54 (DOI 10.5281/zenodo.23418156), where the dark-to-baryon ratio Ω_c/Ω_b = 3√3 was registered against data with the cosmic microwave background (CMB), and it feeds the chain of MM 2.56 (DOI 10.5281/zenodo.23506401) on wide binaries.

In five lines. Standard nucleosynthesis, with the measured deuterium (and helium, in the primary dictionary), constrains ω_b. If ω_c = 3√3·ω_b, the 13 DESI DR2 BAO distances are read with one cosmological parameter fewer than in flat ΛCDM. The fit costs $\Delta\chi^2 = 0.09$ and gives $H_0 = 68.52 \pm 0.59$ km s^{-1} Mpc^{-1}, the known BAO + BBN value. Then $a_0 = cH_0/6 = 1.110\cdot10^{-10}$ m s^{-2} (−0.4σ from SPARC). Read the other way, the BAO alone with 3√3 predict D/H = $(2.40 \pm 0.31)\cdot10^{-5}$, against 2.51 measured (a predictive check, not independent of the ratio test). A test on the first DESI release beyond DR2 is registered here, before those data.

This is a consistency check of one reading (3√3, chosen knowing the data), not a derivation. Its outcome concerns that reading, not the framework.

1. The chain

1. **Deuterium → ω_b.** Standard big-bang nucleosynthesis (BBN) is used as a dictionary: a function of laboratory data (nuclear cross sections, including LUNA's d(p,γ)^{3}He, the neutron lifetime, G, N_eff, T_γ). The primary dictionary uses D and ^{4}He; a deuterium-only dictionary (D-4) is run as a check.
2. **× 3√3 → ω_c.** ω_c = 3√3·ω_b = 5.196·ω_b. In ΛCDM ω_c is free.
3. **→ the sound horizon r_d and H_0.** With ω_b and ω_c fixed, r_d is fixed (CLASS), and the DESI DR2 BAO leave one cosmological parameter free, H_0. ω_b stays a fitted parameter, constrained by the BBN prior.
4. **→ $a_0 = cH_0/6$ → galaxies (SPARC) and wide binaries (the chain of MM 2.56).**

No CMB data enter the fits. Only one Level 2 check (T4, §5) compares the result with Planck afterwards.

2. Data (values read on the sources; SHA-256 of every file in 00_fonti/SHA256_FONTI.txt)

quantity	source	value
BAO distances	DESI DR2 (DESI Collaboration 2025, PRD 112, 083515, arXiv:2503.14738), public Gaussian likelihood files (CobayaSampler/bao_data, commit b7b8a36e; not included,	13 measurements DV/r_d, DM/r_d, DH/r_d, z = 0.295-2.33,

	downloaded by scarica_dati.py with SHA-256 check)	with covariance
primordial D/H	Cooke, Pettini & Steidel 2018 (ApJ 855, 102), eq. 10; PDG 2026 review 24, eq. 24.2 (12 systems)	10^5 D/H = 2.527 ± 0.030; 10^6 D/H = 25.08 ± 0.29
ω_b from BBN (dictionary D-1, primary)	Schöneberg 2024 (arXiv:2401.15054), eq. 5.1: PRyMordial NACRE II with LUNA, marginalised over nuclear rates, PDG 2023 abundances of D **and** ^{4}He; the prior used by DESI DR2 (eq. 14)	0.02218 ± 0.00055
Y_p	PDG 2026 eq. 24.3; Aver et al. 2022 (MNRAS 510, 373) eq. 6	0.245 ± 0.003; 0.2448 ± 0.0033
Li/H observed; predicted	PDG 2026 eq. 24.4; eq. 24.10	$(1.45 \pm 0.25)\cdot10^{-10}$ (strictly a lower bound); $(4.72 \pm 0.7)\cdot10^{-10}$
neutron lifetime	PDG 2026 listings; UCNτ 2025 (PRC 111, 045501); Yue et al. 2013 (PRL 111, 222501); J-PARC 2024 (arXiv:2412.19519)	878.3 ± 0.4 s; 877.82 ± 0.22 +0.20/−0.17 s; 887.7 ± 2.3 s; 877.2 ± 1.7 +4.0/−3.6 s
H_0 for comparison	Planck 2018 VI eq. 14; Riess et al. 2022; Breuval et al. 2024	67.36 ± 0.54; 73.04 ± 1.04; 73.17 ± 0.86
a_0 of SPARC	McGaugh, Lelli & Schombert 2016 (PRL 117, 201101)	g† = 1.20 ± 0.02 (random) ± 0.24 (systematic) $\cdot10^{-10}$ m s^{-2}
BBN tables	PRyMordial (Burns, Tait & Valli 2024, EPJC 84, 86), commit 725d8a8d, file PRyM_Yp_DH_cosmoMC_2023.dat (GPL v3; not included, downloaded by scarica_dati.py)	Y_p and D/H with theoretical errors as functions of ω_b and ΔN_eff

3. Method (Level 1, protocol 02_livello1/PROTOCOLLO_LIVELLO1.md, registered 20 Oct 2026 18:11:01 UTC)

- **Model.** Flat ΛCDM, N_eff = 3.044 (one massive neutrino of 0.06 eV), T_γ = 2.7255 K. r_d = rs_drag of CLASS 3.4.1 on a grid in (ω_b, ω_c), interpolated; distances with an own integrator.
- **Two fits** of $\chi^2 = \chi^2_BAO + ((\omega_b - \omega_b,D)/\sigma_D)^2$ (14 terms):
- ΛCDM: ω_b, ω_c, h free (11 degrees of freedom);
- MM: $\omega_c = 3\sqrt{3}\cdot\omega_b$, with ω_b and h free (12 degrees of freedom).
- **Priors:** flat h ∈ [0.55, 0.85], ω_c ∈ [0.085, 0.155]; ω_b ∈ [0.0196, 0.0249] (edge of the r_d grid, ≥ 4.7σ from the centre of D-1). Posteriors from a Metropolis sampler with 4 chains ($\hat{R} \le 1.0007$ also on split chains, ESS ≥ 11 800).
- **Dictionaries**, all fixed in advance: D-1 primary; D-2 the same with Cooke's D/H; D-3 PDG 2026; D-4 deuterium only (PDG 2026 D/H through the PRyMordial table, with the ln V term of the Gaussian likelihood); D-5 Cooke 2018 before LUNA (historical).
- **Validations, all passed.** The r_d interpolation agrees with CLASS to 0.00003 Mpc, and the distances to $1.3\cdot10^{-6}$. The BAO alone give Ω_m = 0.2975 ± 0.0087 and h·r_d = 101.53 ± 0.74 (DESI eq. 17: 0.2975 ± 0.0086 and 101.54 ± 0.73). ΛCDM + BBN gives H_0 = 68.55 ± 0.60 (DESI eq. 19: 68.51 ± 0.58).

4. Results

quantity	value
χ^2_min: ΛCDM; MM	10.282; 10.369
$\Delta\chi^2$ (one parameter fewer)	**0.087** (registered zone: compatible within 1σ); ΔAIC −1.91, ΔBIC −2.55 (N = 14 by the registered convention)
Ω_c/Ω_b left free (ΛCDM)	5.282 ± 0.228, +0.38σ from 3√3
H_0, MM	**68.52 ± 0.59**; profile $\Delta\chi^2$ = 1: [67.95, 69.05]
H_0, ΛCDM	68.56 ± 0.60
$a_0 = cH_0/6$	**$(1.110 \pm 0.010)\cdot10^{-10}$ m s^{-2}**; against SPARC: z = −0.38 (with H_0 = 73.04: z = −0.07)

Y_p at ω_b,MM (convention Y_P = 4·Y_He of PRyMordial)	0.2468, +0.6σ from PDG and Aver (not independent with D-1)
DESI subsets (MM)	z < 1.1: 68.57 ± 0.64; z > 1.1: 68.32 ± 0.93
deuterium only (D-4)	$\Delta\chi^2$ 0.023; H_0 = 68.72 ± 0.60; Y_p = 0.2469 (+0.6σ, now an independent check)
sensitivity D-1…D-5	$\Delta\chi^2$ 0.02-0.42; H_0 67.99-68.72

The inverse reading. In ΛCDM the BAO alone cannot fix ω_b. With ω_c = 3√3·ω_b they can: ω_b = 0.02316 ± 0.00178 and H_0 = 69.4 ± 1.7, with no BBN input. Through the standard dictionary this predicts **D/H = (2.40 ± 0.31)·10⁻⁵**; the measured value is 2.508 ± 0.029 (z = −0.34). D/H does not enter the BAO fit, but the check shares the BAO and the constraint with the Ω_c/Ω_b test, so it is not independent of it.

Reading of the numbers.

- The chain is consistent: the same BAO quality as ΛCDM with one parameter fewer.
- H_0 is the BAO + BBN value: compatibility, not a new number.
- Today the constraint does not narrow H_0 (0.59 against 0.60), because the data already sit close to it.
- $a_0 = cH_0/6$ is consistent with SPARC but not decisive: the 20 % systematic of SPARC also allows H_0 = 73.
- The wide-binary chain of MM 2.56 uses H_0 = 67.4. The 1.1 km s⁻¹ Mpc⁻¹ difference lies inside the sensitivity range declared there (67.4-73.04). The 2.56 protocol is unchanged.

5. Readings of a premise on time (Level 2)

A premise of the framework treats the standard account of the first minutes as a reading in the present, with one common time for all processes and no objective beginning. Six translations into measured quantities were registered before the computations (protocol of 20 Oct 2026, 18:29:23 UTC, kept outside this archive because it quotes the author's words: see §9). Code and logs are in 03_livello2/. PRyMordial was run with NACRE II rates, LUNA and the full 63-reaction network.

translation	content	D/H	Y_p	Li/H	outcome
T0 common clock	rates of all processes in one time unit: abundances are ratios of counts, the dictionary is unchanged	2.525·10⁻⁵	0.2468 (+0.6σ)	5.11·10⁻¹⁰ = 3.5 × observed	D and He read; Li not read
T1 same everywhere	no significant variation of D/H among 12 absorbers (directions, z_abs 2.4-3.6, metallicities), at the sensitivity of these tests	constancy p = 0.30; slope in z_abs p = 0.61; in metallicity p = 0.17; dipole p = 0.64	—	—	compatible (the dipole is weakly informative: 9 of 12 lines of sight lie between 9h and 16h)
T2a expansion rate = local rate (S = 1)	ΔN_eff free	D + He: ΔN = −0.12 ± 0.21, i.e. S^2 = 0.981 ± 0.034	with BAO and 3√3: ΔN = −0.10 ± 0.18	3.55 ×	compatible with S = 1
T2b hidden mirror sector at the same temperature, with gravitating radiation	ΔN_eq = 6.14 before e^+e^- annihilation, 7.44 after	4.6-5.0·10⁻⁵ (+10 to +11σ)	0.306-0.315 (+20 to +23σ)	2.8-3.1·10⁻¹⁰	**does not hold** for the two equivalent cases examined (the full thermal history of the mirror sector is not simulated); consistent with the assumption of MM 2.44 that hidden radiation does not gravitate
T3 neutron lifetime	bottle 878.3 / beam 888.0 / J-PARC 877.2 s	2.525 / 2.537 / 2.524·10⁻⁵	+0.59 / +1.24 / +0.51σ	3.5 ×	weak hint for the bottle value ($\Delta\chi^2$ = 1.2); He prefers τ_n = 870 ± 15 s

T4 one present baryon-to-photon ratio	ω_b from D/H alone against the CMB	0.02242 ± 0.00056 against 0.02230 ± 0.00021 (z = +0.20)	0.2468	3.5 ×	compatible with the CMB value

Lithium. In every reading that holds, the predicted Li/H is about 3.5 times the value measured in metal-poor halo stars (4.3-4.6σ where the log computes z, with a 15 % theory error). No translation was built on lithium, and none reads it. It stays a measured fact to look at; the PDG treats the stellar value strictly as a lower bound.

What changes for the chain: nothing. The translations that hold leave the dictionary of Level 1 as it is. T2b does not hold in the cases examined; T3 leaves τ_n open, with a weak hint for the value already used.

5b. The same chain with evolving dark energy (check of 20-21 Oct 2026, folder 07_energia_oscura)

Flat w_0w_a (CPL) with DESI DR2 priors; DESI DR2 + BBN, without and with four supernova samples (Pantheon+, Union3, DES-SN5YR, DES Dovekie; the supernova files are not redistributed). The code reproduces DESI's $\Delta\chi^2(w_0w_a - \Lambda\mathrm{CDM})$ without the CMB (−4.90, −10.07, −13.57 against −4.9, −10.1, −13.6).

combination	q free in w_0w_a	$\Delta\chi^2$_q in w_0w_a	$\Delta\chi^2$_q in ΛCDM	H_0 with $3\sqrt{3}$ in w_0w_a
DESI+BBN	5.68 ± 0.48	1.85	0.09	64.7 ± 2.2
+ Union3	5.57 ± 0.51	0.51	0.95	65.8 ± 1.0
+ Pantheon+	5.17 ± 0.56	0.02	1.15	67.1 ± 0.9
+ DES-SN5YR	5.49 ± 0.52	0.31	3.09	66.4 ± 0.8
+ DES Dovekie	5.43 ± 0.54	0.17	1.72	67.0 ± 0.8

- $3\sqrt{3}$ is compatible in all five (registered criterion $\Delta\chi^2$_q ≤ 4). In w_0w_a the error on q doubles, so the check is less stringent.
- In ΛCDM, supernovae push q upwards: the cost of $3\sqrt{3}$ rises to 3.09 with DES-SN5YR. The chain of §1-4 has no supernovae.
- Near the edge $w_0 + w_a \to 0$ the own sound-horizon correction differs from CLASS by up to 0.9 %. The posteriors were rerun with CLASS there (0.00-0.22 % of the samples enter that region; means unchanged within 0.06σ).
- The protocol of this check quotes the author's request and is kept outside the archive (85e5a867…, 20 Oct 20:31:43 UTC). The outcome notes and the corrections after a verification round by Codex are in the folder.

6. Registered prediction (04_previsione/PREVISIONE_CATENA_DEUTERIO_v2.md; time limit of the targets 20 Oct 2026 18:56:04 UTC)

- **Block D (main): DESI beyond DR2 + BBN, no CMB.** This combination is not covered by the prediction of MM 2.54, which uses CMB + BAO.
- **Target:** the first DESI collaboration paper on arXiv after the time limit with BAO beyond DR2 and public Gaussian likelihood files. An arXiv check just before registration found none (04_previsione/controllo_arxiv_v2).
- **Data and chain:** the same chain, frozen; dictionary D-1.
- **Validity:** W1, the BAO-only Ω_m and h·r_d of the paper; W2, its DESI+BBN H_0. The outcome is suspended at prior edges, with non-converged chains or with a negative $\Delta\chi^2$.
- **Rule:** $\Delta\chi^2 \le 4$ «the chain holds»; 4-9 «tension»; > 9 «the reading $\omega_c = 3\sqrt{3}\cdot\omega_b$ does not read BAO + BBN». The block expires on 31 Dec 2028.
- **Block A (secondary): a_0 with independent calibrations.**
- **Target:** the first paper with g† of the radial acceleration relation (McGaugh et al. 2016 function) and a total error ≤ $0.06\cdot10^{-10}$ m s^{-2}.
- **Rule:** z against $a_0 = cH_0/6$ of the chain: $|z| \le 2$ «holds», 2-3 «tension», > 3 «does not hold». The block expires on 31 Dec 2030.
- **Addendum 1** (04_previsione/AGGIUNTA_1_PREVISIONE_w0wa.md, registered 21 Oct 2026 00:29:34 UTC, corrected at 00:32:33 UTC, both before any DESI data beyond DR2; arXiv checked). The rule of block D does not change. Next to its outcome, the same computation is reported in w_0w_a, with a qualifier fixed in advance:

- if block D gives «tension» or worse, $\Delta\chi^2$_q in w_0w_a ≤ 4 and the same data prefer w_0w_a by at least 2σ ($\Delta\chi^2 \leq -6.18$), the text says «the outcome of block D depends on the background model»;
- the qualifier is not given when 3√3 is saved only by dark energy that matters before the drag epoch ($w_0 + w_a > -0.3$). A test showed that this region can absorb a wrong q (synthetic data with q = 6.5: $\Delta\chi^2$ 91 in ΛCDM, 0.63 in w_0w_a). Program decidi_aggiunta_w0wa_v2.py, 14 tests passed.
- **Program and tests.** decidi_previsione_deuterio_v2.py decides; its tests include a dry run on DR2 that reproduces the numbers above and the counterexamples of the second verification round. All pass. One registered test was faulty: it altered a zero element, so it never tested asymmetry (21 of 22 passed). It was redone with a non-zero element and passes (ESITO_PROVE_v2.md).

7. What is known and what is specific to this analysis

- **Known.** BAO + BBN → H_0 without the CMB: Addison et al. 2018 (66.98 ± 1.18), Cuceu et al. 2019 (67.6 ± 1.1), Schöneberg, Lesgourgues & Hooper 2019, Schöneberg et al. 2022 (67.6 ± 1.0; 68.3 ± 0.7 with ShapeFit), DESI DR2 (68.51 ± 0.58). Krolewski, Percival & Woodfinden 2024 (arXiv:2403.19227) obtain H_0 from BBN with a baryon fraction *measured* in galaxy clustering, by a different route and with a wider error (67.1 +6.3/−5.3). Our H_0 is the BAO + BBN value of the first group of analyses.
- **Specific to this analysis.** The ratio is imposed, q = Ω_c/Ω_b = 3√3, instead of measured: one parameter fewer. Two more pieces come with it: the inverse reading (BAO → deuterium), and the link to $a_0 = cH_0/6$ and to the wide-binary chain. A targeted search found no analysis with this constraint together with BAO and BBN; it was not a complete review.
- **Not claimed.** 3√3 is not derived: it is a reading chosen knowing the data (grade B*). With the CMB, 3√3 lies about 2σ from Planck (MM 2.44 and 2.54); that comparison is outside this chain. Verlinde's emergent gravity gives the same ratio for Ω_b = 4/81 (MM 2.54).

8. How this can turn out wrong

- Block D gives $\Delta\chi^2 > 9$ with DESI beyond DR2: the reading ω_c = 3√3·ω_b would then not read BAO + BBN.
- A precise g† far from $cH_0/6$ (block A).
- A shift of the D/H average, or of the d(p,γ)³He and d+d rates, beyond the D-1 errors. This moves ω_b and therefore the whole chain; D-5 shows the size of the effect of pre-LUNA rates.

9. Verification, registration and files outside the archive

- **Registration.** Every protocol and program was registered with SHA-256 and UTC time before it was run. The registers are the REGISTRO.txt files in each folder. Registered files were never edited; corrections are separate files.
- **Two verification rounds by Codex (GPT-6, model gpt-6-luna).**
- Round 1, on Level 1: «not ready», 11 points. Ten were applied (02_livello1/CORREZIONI_LIVELLO1_1.md). One was not accepted, with the source: Y_p is compared in PRyMordial's convention Y_P = 4·Y_He (arXiv:2307.07061 eq. 2.20).
- Round 2, on everything: «not ready», 16 points. All were applied (05_astra_giro2/CORREZIONI_GIRO2.md), mostly robustness of the decision program and the scope of T2a (it tests the expansion rate, not G separately).
- No third round on the science.
- **A check of this package** (texts, run_all.py, build script): «not ready», 11 points, all applied: wording of the public texts and a stricter run_all.py; no number changed (verifica_pacchetto/). No further round.
- **Kept outside the archive because they quote the author's words** (SHA-256 and registration time in the registers; available on request). The build script excludes them by name and checks every packed file for a list of the author's phrases. Quotation marks in the correction files quote earlier texts by Claude, not the author.
- the premise;
- the Level 2 protocol (b3abd2a0…, 18:29:23 UTC);
- the Level 2 outcome notes (0d7506a9…, d1dbabe1…);
- the Level 0 inventory v1 (a40802fd…);
- the protocol of the w_0w_a check (85e5a867…);
- the messages sent to the verifier.

10. Reproduction

- **Requirements:** Python ≥ 3.10 with numpy and scipy. For the full run also classy (CLASS 3.4.1) and the PRyMordial code.
- `python scarica_dati.py` downloads the DESI DR2 likelihood files (commit b7b8a36e of

CobayaSampler/bao_data) and PRyMordial (commit 725d8a8d), and checks their SHA-256 (DESI 9ac154ab…, 252a1432…; BBN table bf4d1685…).
- `python run_all.py` reruns the validations, Level 1, T1, the Level 2 comparison and the tests of the decision program, and compares every output with the reference logs (about 12 minutes). A program that exits with an error, a test marked [NO] (other than the faulty registered test A12, redone separately) or a number that differs by more than three units of its last printed digit is reported as DIFFERENT. The two PRyMordial runs are compared on their JSON output with a relative tolerance of 0.5 %: PRyMordial's integration differs between Windows and Linux by up to 0.3 % in D/H and Li/H (verifica_pacchetto/confronto_piattaforme_bbn_linux.log), far below the physical errors.
- The addendum tests (prove_aggiunta_w0wa_v2.py) are part of the default run. The w_0w_a check with supernovae (07_energia_oscura) needs the four supernova files, which are not redistributed (SHA-256 of the pack in 00_fonti/SHA256_FONTI.txt); without supernovae it runs from the archive.
- `python run_all.py --completo` also reruns the PRyMordial runs, T2a, the inverse reading and the r_d grid (about 25 minutes more, single core).
- The working texts (protocols, notes, corrections) are in Italian; this README summarises them.

The author is not a professional physicist. The computations were done with AI assistants (Claude, with two checks by Codex) and have not been reviewed by a human referee. Texts and materials: CC BY 4.0. Code: MIT. Third-party data and code are not redistributed (the DESI files carry no explicit licence; PRyMordial is GPL v3): scarica_dati.py downloads them.

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